DAA ASSIGNMENT

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Quick Sort by taking three different pivots and Quick select

CODE: -

```
#include <stdio.h>
void swap(int *a, int *b){
    int t = *a;
    *a = *b;
    *b = t;
}
int median(int a[], int l, int r){
    int m = (l + r) / 2;
    if(a[l] > a[m]) swap(&a[l], &a[m]);
    if(a[l] > a[r]) swap(&a[l], &a[r]);
    if(a[m] > a[r]) swap(&a[m], &a[r]);
    swap(&a[m], &a[r-1]);
    return a[r-1];
}
int partition(int a[], int l, int r){
    int pivot = median(a, l, r);
    int i = l, j = r - 1;
    while(1){
        while(a[++i] < pivot);
        while(a[--j] > pivot);
        if(i < j) swap(&a[i], &a[j]);
        else break;
    }
    swap(&a[i], &a[r-1]);
    return i;
}
```

```
void quicksort(int a[], int l, int r){
    if(l < r){
        int p = partition(a, l, r);
        quicksort(a, l, p - 1);
        quicksort(a, p + 1, r);
    }
}

int quickselect(int a[], int l, int r, int k){
    if(l == r) return a[l];

    int p = partition(a, l, r);

    if(k == p) return a[p];
    else if(k < p) return quickselect(a, l, p - 1, k);
    else return quickselect(a, p + 1, r, k);
}

int main(){
    int n, k;
    printf("Enter number of elements: ");
    scanf("%d", &n);
    int a[n];
    printf("Enter elements:\n");
    for(int i = 0; i < n; i++)
        scanf("%d", &a[i]);
    quicksort(a, 0, n - 1);
    printf("Sorted array:\n");
    for(int i = 0; i < n; i++)
        printf("%d ", a[i]);
    printf("\nEnter k value for kth smallest element: ");
    scanf("%d", &k);
    printf("kth smallest element = %d", quickselect(a, 0, n - 1, k - 1));
}
```

OUTPUT: -

```
┌──(gnaneswar993㉿kali)-[~]
└─$ nano quickselect_sort.c

┌──(gnaneswar993㉿kali)-[~]
└─$ gcc quickselect_sort.c -o quickselect_sort

┌──(gnaneswar993㉿kali)-[~]
└─$ ./quickselect_sort
Enter number of elements: ^[[A
Enter elements:
Sorted array:

zsh: segmentation fault  ./quickselect_sort

┌──(gnaneswar993㉿kali)-[~]
└─$ ./quickselect_sort
Enter number of elements: 8
Enter elements:
5
9
7
6
4
6
2
0
Sorted array:
0 2 4 5 6 6 9 7
Enter k value for kth smallest element: 5
kth smallest element = 6

┌──(gnaneswar993㉿kali)-[~]
```